Alexander Rein, B.S.

Denver, Colorado

Background

- Analytical Chemistry
- Research & Development
- GC, GC/MS, ICP-MS,
- Wavelength Dispersive X-Ray Fluorescence
- UV, Vis Spectroscopy and FT-IR
- HPLC
- Data collection and analysis
- Laboratory analysis
- Chemical inventory
- Particle sizing/distribution
- Pore volume measurement
- Flow analysis (powder and fines)
- Organic synthesis
- Environmental law
- Statistical analysis/data interpretation
- Environmental health and safety
- Instrumental analysis
- Microbiological procedures
- Water treatment techniques
- Field sampling and analysis

EDUCATION

Metropolitan State University of Denver,
B.S. Environmental Science: concentration in Environmental Chemistry

EMPLOYMENT HISTORY

Weatherford Laboratories
Core Evaluation Specialist-Rock Mechanics
October 2014 - Present

- Hydraulic fracture design
- P&S wave velocities
- Pore volume compressibility
- Well bore stability
- Mohr Coulomb failure analysis
- Proppant Embedment testing
- Triaxial strength testing (single and multi-stage)
- Unconfined compressional strength testing
- Field application development

ADA Carbon Solutions
R&D Laboratory Technician
May 2014-October 2014

- Conduct proprietary laboratory analyses.
- Developed and maintained sample management system.
- Reconciliation of chemical inventory and MSDS library for right to know compliance.
- Maintaining stock of laboratory supplies and equipment.
- Data analysis and database control.
- New product development including scale up procedures.
- Conducted gauge R&R testing to determine operational quality

Accutest Laboratories
Laboratory Analyst
January 2014 – May 2014

- Prep and Conduct 8015/8021 GRO/BTEX analysis of environmental samples via GC, GC/MS
- Data editing and approval within the Laboratory Information Management System.
- Manual peak integrations
- Report preparation
- Organic extractions
- Maintaining laboratory instruments

National Renewable Energy Laboratory
Laboratory Tester
January 2014 - February 2014

- Determination of total solids in algal biomass
- Determination of total carbohydrates in algal biomass
- Determination of lipids in algal biomass as fatty acid methyl esters (FAME analysis)
- Instrumental analysis via HPLC, UV Vis Spectroscopy, HPAEC-PAD, GC-MS, TOF/MS-ESI
- Preparation of continuing calibration verification standards

Core Laboratories, Inc.
Core Analyst
May 2013 - November 2013

- Sampled & analyzed material from unconventional oil & gas reservoirs
- Performed routine tests to determine samples chemical or physical characteristics
- Performed all defined analytical standards and procedures including adsorption isotherms.
- Set up, adjusted and operated lab equipment
- Performed basic analytical tests as related to oil exploration
- Compiled and reviewed raw data sheets
- Onsite core analysis and processing
- Field mobilization
- Inventory control

Metropolitan State University of Denver, Chemistry Department
Laboratory Technician
August 2012 - May 2013

- Prepared and maintained laboratories for student use
- Assisted in teaching Organic Chemistry I Lab
- Prepared solutions used in experiments for all course levels
- Collected, disposed of, and neutralized aqueous and organic waste products
- Calibration and Optimization of analytical instruments
- Maintained and enforcing all safety protocol

The Brand Consortium
Public Relations Assistant
January 2008 - May 2013

- Assisted company operations in public relations, marketing campaigns and client management
- Handled local ongoing operations and events

Construction Apprentice
Holiday Home Builders
August 2005 - January 2008

- Assisted the foreman in daily inspections, janitorial work, landscaping, appliance installation and repair

RESEARCH EXPERIENCE

Environmental Field Studies
Chemical Fractional Analysis of High Alpine Soils: Rocky Mountain National Park

This paper investigated the concentrations of phosphates and phosphorous in various fractions of soil, in a variety of soil horizons. A sequential extraction method involving decreasingly mild solvents was used to extract phosphorous from different components of soil with differing levels of adhesion via chemisorption. The extracts were tested in a variety of fashions including the use of a colorimeter to simulate use of field equipment (LaMotte Smart Lab 3). The laboratory techniques used to analyze samples required Inductively Coupled Plasma Mass Spectrometry and Atomic Absorbance Spectrometry. *(Continued on page 3)*

This data was used to determine the amounts of phosphorous made biologically available in response to freeze thaw events. The adjacent watersheds were determined to be at risk of eutrophication due to the elevated concentration of phosphates. It remains unknown where the phosphates and phosphorous came from in this high alpine region, as it has not seen a great deal of human activity, which is commonly the cause of these occurrences.

Advanced Organic Synthesis
Selective Nuclear Mono-bromination of Resorcinol

This research investigated the silica gel catalyzed reaction of 1,3-bezenediol and N-bromosuccinimide. The reaction protocol was altered in several fashions causing changes in selectivity. Poly-bromination is a common problem with electrophilic aromatic substitution with respect to phenolic compounds. It was found that the reaction procedure cited in literature functioned better under altered solvent conditions and excess quantities of resorcinol. Attempts to alter the selectivity of resorcinol included alterations of the solvent, reactant quantities, reaction temperature, and finally by deactivating the aromatic ring through esterification of the hydroxyl groups.

PROFESSIONAL MEMBERSHIPS/ACTIVITES

- American Chemical Society
- American Association of Petroleum Geologists
- Council on Undergraduate Research
- Metropolitan State University of Denver Chemistry Club
- Metropolitan State University of Denver Earth and Atmospheric Science Club